Checkpoint 4 — Newton–Raphson Update for Bayesian Logistic Regression

INFO 521 — Machine Learning Foundations

> **i** Course-schedule placement
>
> Sitting: **Week 5** (with Module 5 — Approximate Inference & Classification) · retake: Week 6 · gates **Project 2 · Gate A**.

Gates: Project 2.1 (Approximate Inference) — and with it, Gate A. **Format:** in-class, closed-book, ~15–20 min, one derivation. Graded Satisfactory / Not-Yet. **Retakeable** (an alternate version is provided below).

This checkpoint tests the study-guide outcome *"derive a second-order optimizer for a non-conjugate posterior."* It is the supervised-gate counterpart to Checkpoints 1–2: where those derived the closed-form least-squares/MLE solution, this one derives the **iterative** update you need once conjugacy is gone.

Prompt (Version A)

Bayesian logistic regression places a Gaussian prior $\mathbf{w} \sim \mathcal{N}(\mathbf{0}, \mathbf{S}_0)$ on the weights, with a Bernoulli likelihood $y_n \sim \mathrm{Bernoulli}\,(\mu_n)$, $\mu_n = \sigma(\mathbf{w}^\top \mathbf{x}_n)$ and σ the logistic sigmoid. Writing $E(\mathbf{w}) = -\log p(\mathbf{w} \mid \mathcal{D})$ (up to a constant), derive (i) the **gradient** ∇E, (ii) the **Hessian** $\nabla^2 E$, and (iii) the **Newton–Raphson update** for the MAP estimate. State why the update is called *iteratively reweighted least squares*.

Prompt (Version B — retake)

Explain the **Laplace approximation** of the posterior in this same model: what it is, *why* the **negative Hessian of the log-posterior at the mode** is the posterior **precision**, and give **one** situation in which the approximation is poor.

Satisfactory criteria

- (A) Correct ∇E and $\nabla^2 E$, including the prior term $\mathbf{S}_0^{-1}\mathbf{w}$ (gradient) and $\mathbf{S}_0^{-1}$ (Hessian).
- (A) States the update $\mathbf{w} \leftarrow \mathbf{w} - \mathbf{H}^{-1}\mathbf{g}$ and identifies the reweighting matrix $\mathbf{R} = \mathrm{diag}\,(\mu_n(1-\mu_n))$ as the "weights" that make it reweighted least squares.
- (B) Names the Gaussian centred at the mode; argues the precision = curvature of the log-posterior at the mode; gives one valid failure case (skewed / multimodal / near-separable posterior).

Answer key

Let $\boldsymbol{\mu} = (\mu_1, \ldots, \mu_N)^\top$, $\mathbf{R} = \mathrm{diag}\left(\,\mu_n(1-\mu_n)\,\right)$, and $\mathbf{X}$ the design matrix with rows $\mathbf{x}_n^\top$.

A:

$$\nabla E = \mathbf{X}^\top(\boldsymbol{\mu} - \mathbf{y}) + \mathbf{S}_0^{-1}\mathbf{w}, \qquad \nabla^2 E = \mathbf{X}^\top \mathbf{R}\,\mathbf{X} + \mathbf{S}_0^{-1}.$$

$$\mathbf{w} \leftarrow \mathbf{w} - \left(\, \mathbf{X}^\top \mathbf{R}\,\mathbf{X} + \mathbf{S}_0^{-1}\right)^{-1} \left(\, \mathbf{X}^\top(\boldsymbol{\mu} - \mathbf{y}) + \mathbf{S}_0^{-1}\mathbf{w}\,\right).$$

It is *reweighted least squares* because each Newton step solves a weighted least-squares problem with weights $\mathbf{R}$ that are **recomputed** from $\boldsymbol{\mu}$ every iteration.

B: Taylor-expand E to second order at the mode $\mathbf{w}_{\mathrm{MAP}}$. The first-order term vanishes ($\nabla E = \mathbf{0}$ at the mode), leaving $E(\mathbf{w}) \approx E(\mathbf{w}_{\mathrm{MAP}}) + \frac{1}{2}(\mathbf{w} - \mathbf{w}_{\mathrm{MAP}})^\top \mathbf{H}\,(\mathbf{w} - \mathbf{w}_{\mathrm{MAP}})$ with $\mathbf{H} = \nabla^2 E(\mathbf{w}_{\mathrm{MAP}})$. Exponentiating gives a Gaussian $q(\mathbf{w}) = \mathcal{N}(\mathbf{w}_{\mathrm{MAP}}, \mathbf{H}^{-1})$: since a Gaussian's log-density has curvature equal to its **precision**, matching the curvature of the log-posterior at the mode sets the precision to $\mathbf{H} = \mathbf{X}^\top \mathbf{R}\,\mathbf{X} + \mathbf{S}_0^{-1}$ (the negative Hessian of the *log*-posterior). It is poor when the true posterior is far from Gaussian — e.g. skewed, multimodal, or near-separable data (few positives, or perfectly separable classes), where a single mode-centred Gaussian misstates the spread.